

CS6886: Systems for Deep Learning

Assignment 4: ANN-to-SNN Conversion

“If you optimize everything, you will always be unhappy.”
- Donald Knuth

In this assignment, we will optimize everything in ANN conversion: accuracy, sparsity and energy.

Extract the Assignment_4.zip archive, where the intended root folder contains a setup.py file and the iPython notebook you will be working in for this assignment.

Install SpikingJelly using the the setup.py file with the command

```
python setup.py install
```

Do not install with pip as we will be using a slightly modified fork of the library. Prerequisites for this assignment are an iPython notebook environment like Jupyter, and PyTorch ([install the GPU version](#) if you want to utilise CUDA).

All steps for the assignment are discussed in the notebook. For the assignment tasks, you can only change/add to the code in the iPython notebook, and in the files cnn_mnist.py and loader.py from the folder spikingjelly/activation_based/ann2snn/examples. You can add code however you like for stretch goals as long as you're using PyTorch and SpikingJelly only for the ANN and SNN logic respectively.

Summarise and explain your process through the assignment either in the iPython notebook or in a companion PDF submission.

Rename the notebook to <your roll number>_6886A4.ipynb.

Before submitting your iPython notebook, please “Restart and run all cells”. There will be penalties if cell outputs in your notebook are manipulated, not in order, or unreproducible.

[Link to the archive](#).